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Ans-5) In RAG, documents are cut into chunks, turned into embedding vectors and saved in a vector store. The question is embedded, the nearest chunks are fetched by cosine similarity and pasted into the prompt, and the LLM answers from them. Problem: the answer is not in the top-k chunks because of vocabulary mismatch -> add BM25 keyword search and a re-ranker. Problem: the model ignores the context and invents facts -> instruct it to answer only from the passages and to say it does not know.

2) Queries, keys and values are produced from each token's embedding with three learned projection matrices. Scores are $q_i \cdot k_j$ for all j (QK^T), divided by $\sqrt{d_k}$, then softmax turns each row into weights adding up to 1. The output is the weighted sum of the values. Without the scaling, large d_k makes the dot products large, the softmax saturates and the gradients vanish.

Q.1 A model overfits when it fits the noise of the training set and so does well on training data but badly on new data. Weight decay (L2) keeps weights small; early stopping halts training when the validation loss rises.

(3) Precision = 40/(40+10) = 0.80; Recall = 40/(40+20) = 0.67; F1 = 2(0.8)(0.67)/(1.47) = 0.73; Accuracy = (40+30)/100 = 0.70

4. GANs train a generator against a discriminator; diffusion models learn to remove Gaussian noise step by step. GANs are hard to train and may mode-collapse; diffusion is stable and diverse. GANs sample in one pass, diffusion needs many steps.